

Total No. of Questions : 8]

SEAT No. :

P272

[6003]-350

[Total No. of Pages : 2

T.E. (Computer)

**INTERNET OF THINGS AND EMBEDDED SYSTEMS
(2019 Pattern) (Semester - I) (Elective-I) (310245(A))**

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Attempt Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Neat diagram must be drawn wherever necessary.
- 3) Assume suitable data, if necessary.

- Q1)** a) Demonstrate the use of different networking components and devices required for the IoT application design. Consider smart irrigation system as an example for it. **[6]**
- b) Demonstrate the IoT communication Models. **[6]**
- c) Illustrate the different pillars of IoT. **[6]**

OR

- Q2)** a) Illustrate steps of IoT design methodology for smart forest fire detection. **[6]**
- b) Demonstrate the Web socket API with suitable IoT system. **[6]**
- c) Categorize requirement of connectivity technologies for IoT system development and explain any one of them in brief. **[6]**

- Q3)** a) Examine the different issues in standardization of IoT Protocols. **[6]**
- b) Classify the different IoT Protocols used at Network layer and explain any one of them in brief. **[6]**
- c) Show the use of LoRa protocol in the smart irrigation system development. **[5]**

OR

- Q4)** a) Classify between M2M and SCADA Protocol. **[6]**
- b) Demonstrate the use of IP based protocols in the IoT Applications. **[6]**
- c) Apply the appropriate IoT protocol to develop smart irrigation system with proper explanation. **[5]**

P.T.O.

Q5) a) Examine how Cloud Computing is an IoT enabling technology with the suitable example. [8]

b) Use the knowledge of Cloud Computing to demonstrate.

i) Auto Bahn for IoT

ii) Xively Cloud for IoT

[10]

OR

Q6) a) Show that Cloud computing is the fusion of Grid Computing and SOA. [8]

b) Apply the concept of cloud computing to design the smart home system with proper explanation. [10]

Q7) a) Predict the possible vulnerabilities in designing smart home intrusion detection system. [8]

b) Apply the key elements of IoT security for securing the forest fire detection system with proper explanation. Enlist possible threats may encountered in designing such applications. [9]

OR

Q8) a) Illustrate the challenges in securing IoT applications. [8]

b) Use security concepts to identify different misuse cases (at least 03) in each of the following IoT applications: [9]

i) Smart Home Automation.

ii) Smart Parking System.

iii) Smart Irrigation Sytem.



Total No. of Questions : 8]

SEAT No. :

P-7541

[Total No. of Pages : 2

[6180]-49

T.E. (Computer Engineering)

**INTERNET OF THINGS AND EMBEDDED SYSTEMS
(2019 Pattern) (Semester - I) (Elective - I) (310245(A))**

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Attempt Q1 or Q2, Q3 or Q4, Q5 or Q6, Q7 or Q8.
- 2) Neat diagram must be drawn wherever necessary.
- 3) Assume suitable data if necessary.

- Q1)** a) Explain the steps involved in the IoT design methodology. [6]
b) Explain the concept of Machine-to-Machine (M2M) communication in the context of IoT. [6]
c) Illustrate the different pillars of IoT. [6]

OR

- Q2)** a) Identify and explain the key components of an IoT network architecture. [6]
b) Demonstrate the Web socket API with suitable IoT system. [6]
c) Discuss the advantages and limitations of using IoT communication APIs in home automation systems. [6]

- Q3)** a) Analyze the characteristics and functionalities of M2M protocols used in IoT applications. [6]
b) Analyse the Modbus protocol and its usage in industrial IoT applications. Discuss the features and functionalities of Modbus, its communication modes, and the benefits it offers in connecting devices in industrial automation. [6]
c) Analyse the working principles and applications of the RFID protocol in IoT system. [5]

P.T.O.

OR

- Q4)** a) Classify between M2M and SCADA Protocol with proper example. [6]
b) Demonstrate the use of IP based protocols in the IoT Applications. [6]
c) Show the use of LoRa protocol in the smart irrigation system development. [5]

- Q5)** a) Examine how Cloud Computing is an IoT enabling technology with the suitable applications. [8]
b) Design a cloud storage model for an IoT-based healthcare application. Consider the storage requirements, data security, and privacy concerns associated with sensitive patient health records. Discuss the pros and cons of using public, private, and hybrid cloud storage options. [10]

OR

- Q6)** a) Show that Cloud computing is the fusion of Grid Computing and SOA. [8]
b) Design a home automation system using the AutoBahn for IoT and Xively Cloud for IoT communication APIs. Discuss how these APIs can be used to enable device control, data collection, and remote monitoring of various home appliances and sensors. [10]

- Q7)** a) Design an introduction to IoT security, highlighting the unique challenges and vulnerabilities associated with IoT deployment. [8]
b) Illustrate the challenges in securing IoT applications. [9]

OR

- Q8)** a) Predict the possible vulnerabilities in designing smart home intrusion detection system. [8]
b) Design a case study on designing a secure IoT home intrusion detection system. Identify the challenges and considerations involved in ensuring the confidentiality, integrity, and availability of data, as well as the timely detection and response to potential security breaches. [9]



Total No. of Questions : 8]

SEAT No. :

PB3781

[6262] 39

[Total No. of Pages : 2

T.E. (Computer Engineering)
INTERNET OF THINGS & EMBEDDED SYSTEMS
(2019 Pattern) (Semester-I) (310245 A) (Elective- I)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6 and Q.7 or Q.8.
- 2) Draw neat & labelled diagrams if necessary.
- 3) Figures to the right indicate full marks.
- 4) Assume Suitable data, if necessary.

- Q1)** a) Discuss the various steps in IoT design methodology? [6]
b) Demonstrate the use of RFID with the help of suitable IoT application. [6]
c) Explain M2M communication in detail. [5]

OR

- Q2)** a) Describe device and component integration for IoT based home automation system. [6]
b) Demonstrate the working of push-pull communication model using diagram with suitable application. [6]
c) What are horizontal and vertical of IoT applications? [5]

- Q3)** a) What is 6LoWPAN? 6LoWPAN and EPC standardization. [6]
b) Explain SCADA protocol standardization. [6]
c) What is MQTT and explain in detail. [6]

OR

- Q4)** a) Explain LoRA based smart Irrigation system. [6]
b) Explain MODBUS protocol in detail. [6]
c) Examine that why ZigBee is popular than Wi-Fi and Bluetooth in IoT. [6]

P.T.O.

- Q5)** a) Define software define networking & Explain architecture of SDN. [6]
b) Write a short note on cloud standardization. [6]
c) Describe the IoT messaging mechanisms called WAMP (Auto Bahn for IoT). [6]

OR

- Q6)** a) Define Cloud of Things & What is cloud communication API? [6]
b) Explain the Python Web application framework Django. [6]
c) Explain the different cloud-based services offered by Amazon for IoT. [6]

- Q7)** a) Write a short note on Light weight cryptography. [6]
b) What is Activity Modelling of Threats? & Explain access control issue with respect to IoT security. [6]
c) What are the different vulnerabilities of IoT and how to handle? [5]

OR

- Q8)** a) Examine how threat model is useful in securing IoT applications. [6]
b) List out security requirements for IoT base systems. & Discuss some vulnerabilities in IoT. [6]
c) What are the challenges in designing the IoT application. [5]



Total No. of Questions : 8]

SEAT No. :

PC1722

[Total No. of Pages : 2

[6353]-39

T.E. (Computer Engineering)

**INTERNET OF THINGS AND EMBEDDED SYSTEMS
(2019 Pattern) (Semester - I) (310245 (A)) (Elective -I)**

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Attempt Q1 or Q2, Q3 or Q4, Q5 or Q6, Q7 or Q8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Assume suitable data if necessary.

- Q1)** a) Illustrate the working of any 03 Communication models with suitable application. [6]
- b) Demonstrate SCADA pillar of IoT with Suitable IoT System. [6]
- c) Illustrate connectivity technologies used in IoT with proper example. [6]

OR

- Q2)** a) Illustrate step wise IoT design methodology. [6]
- b) Demonstrate the typical RFID system with the help of suitable IoT Application. [6]
- c) Categorize different IoT design methodologies for Smart Irrigation system. [6]

- Q3)** a) Illustrate the use of MQTT protocol with suitable IoT Application. [6]
- b) Classify the different Topology of IEEE 802. 15.4 and explain with suitable diagram. [6]
- c) Show the use of LoRa protocol in any suitable IoT application development. [5]

OR

- Q4)** a) Categorize between SCADA and WSN protocols. [6]
- b) Illustrate the various IoT applications developed using IP based protocols. [6]
- c) Illustrate use of Zigbee in the smart home system. [5]

P.T.O.

Q5) a) Demonstrate Amazon Cloud platform usage for IoT applications. [8]

b) Use the knowledge of Cloud Computing to demonstrate. [10]

i) Autobahn for IoT

ii) Xively Cloud

OR

Q6) a) Show that WAMP and its key concepts are useful in Cloud based IoT application Development. [8]

b) Apply the concept of cloud computing to design the Weather forecasting system with proper explanation. [10]

Q7) a) Predict the possible security challenges in designing secure IoT applications. [8]

b) Illustrate the how the classic pillars of information assurance useful in securing the IoT application. [9]

OR

Q8) a) Illustrate the threat model is playing role in securing IoT applications.[8]

b) Demonstrate the security requirements of IoT Applications. [9]



Total No. of Questions : 8]

SEAT No. :

PD4245

[6403]-39

[Total No. of Pages : 2

T.E. (Computer Engineering)

INTERNET OF THINGS AND EMBEDDED SYSTEMS

(2019 Pattern) (Semester - V) (310245(A)) (Elective - I)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Attempt Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Neat diagram must be drawn wherever necessary.
- 3) Assume suitable data if necessary.

- Q1)** a) Evaluate the importance of each step in the IoT design methodology.[6]
- b) Demonstrate SCADA pillar of IoT with Suitable IoT System. [6]
- c) Illustrate connectivity technologies used in IoT with proper example.[6]

OR

- Q2)** a) Discuss the suitability of different communication models for specific IoT applications. [6]
- b) Demonstrate the use of RFID with the help of suitable IoT Application.[6]
- c) Provide a case study of a home automation system using IoT communication models. [6]

- Q3)** a) Critically evaluate the role of wireless sensor network (WSN) protocols in enabling efficient communication among sensor nodes in IoT deployments. [6]
- b) Classify the different Topology of IEEE 802.15.4 and explain with suitable diagram. [6]
- c) Analyze the Modbus protocol and its usage in industrial IoT applications.[5]

OR

P.T.O.

- Q4)** a) Analyze the role of RFID protocols in supply chain management and inventory tracking. [6]
- b) Illustrate the various IoT applications developed using IP based protocols. [6]
- c) Critically evaluate the advantages and limitations of Zigbee technology in home automation systems. [5]

- Q5)** a) Design a Django-based RESTful API for an IoT system. [8]
- b) Design a cloud storage model for a fleet management system. Discuss the data synchronization, real-time data processing, and backup strategies to ensure seamless access to vehicle telemetry and operational data from multiple locations. [10]

OR

- Q6)** a) Demonstrate Amazon Cloud platform usage for IoT applications. [8]
- b) Apply the concept of cloud computing to design the Weather forecasting system with proper explanation. [10]
- Q7)** a) Predict the possible security challenges in designing secure IoT applications. [8]
- b) Illustrate the challenges in securing IoT applications. [9]

OR

- Q8)** a) Examine how threat model is useful in securing IoT applications. [8]
- b) Design a security model for an IoT ecosystem, encompassing all the necessary components, protocols and mechanisms to ensure end-to-end security. [9]

x

x

x

Total No. of Questions : 8]

SEAT No. :

PA-1445

[Total No. of Pages : 2

[5926]-61

T.E. (Computer Engineering)

**INTERNET OF THINGS AND EMBEDDED SYSTEMS
(2019 Pattern) (Semester - I) (Elective - I) (310245 A)**

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Neat diagram must be drawn wherever necessary.
- 3) Assume suitable data, if necessary.

- Q1)** a) Demonstrate the working of push-pull Communication model using Diagram with suitable application. [6]
- b) Illustrate any Communication API with Suitable IoT System. [6]
- c) Examine the use of each pillar of IoT with proper example. [6]

OR

- Q2)** a) Illustrate steps of IoT design methodology for weather forecasting system. [6]
- b) Demonstrate the use of RFID with the help of suitable IoT Application. [6]
- c) Classify different connectivity technologies required for IoT system development and explain any one of them in brief. [6]

- Q3)** a) Demonstrate the need of standardization of IoT Protocols. [6]
- b) Classify the different Topology of IEEE 802.15.4 with proper applications. [6]
- c) Show the use of LoRa protocol in suitable IoT application development. [5]

OR

- Q4)** a) Show the merits and demerits between RFID and SCADA protocol. [6]
- b) Illustrate the various IoT applications developed using IP protocols. [6]
- c) Examine that why ZigBee is popular than Wi-Fi and Bluetooth in IoT. [5]

P.T.O.

- Q5)** a) Demonstrate the Django framework with the suitable supporting application. [8]
b) Use the knowledge of Cloud computing to demonstrate need of [10]
i) Amazon Auto Scaling
ii) Xively Cloud for IoT.

OR

- Q6)** a) Show how WAMP, its related concepts are useful in Cloud based IoT application Development. [8]
b) Apply the concept of cloud computing to design the smart home system with proper explanation. [10]
- Q7)** a) Demonstrate the possible challenges in designing secure IoT applications. [8]
b) Show the use of classic pillars of information assurance while securing the IoT application. [9]

OR

- Q8)** a) Examine how threat model is useful in securing IoT applications. [8]
b) Use security concepts to identify different threats (at least 03 in each) in the following IoT applications: [9]
i) Smart irrigation
ii) Smart home System
iii) Smart Surveillance System

Total No. of Questions : 8]

SEAT No. :

P809

[5870] - 1129

[Total No. of Pages : 2

T.E. (Computer Engineering)

**INTERNET OF THINGS AND EMBEDDED SYSTEMS
(2019 Pattern) (Semester - I)(Elective - I) (310245A)**

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q. 1 or Q. 2, Q. 3 or Q. 4, Q. 5 or Q. 6, Q. 7 or Q. 8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Assume suitable data, if necessary.

Q1) a) Demonstrate the working of Publish-Subscribe Communication model using Diagram with suitable application. [6]

b) Illustrate REST based Communication API with Suitable IoT System. [6]

c) Classify the four pillars of IoT. [6]

OR

Q2) a) Illustrate steps of IoT design methodology for smart irrigation system. [6]

b) Demonstrate the use of SCADA with the help of suitable IoT Application. [6]

c) Categorize different connectivity technologies required for IoT system development and explain any one of them in brief. [6]

Q3) a) Illustrate different issues with standardization of IoT Protocols. [6]

b) Classify the different Topology of IEEE 802.15.4 and explain with suitable diagram. [6]

c) Show the use of LoRa protocol in any suitable IoT application development. [5]

OR

P.T.O.

- Q4)** a) Classify between RFID and SCADA Protocol. [6]
b) Illustrate the various IoT applications developed using IP based protocols. [6]
c) Show with suitable reasons why Zigbee is popular than Wi - Fi and Bluetooth in IoT. [5]
- Q5)** a) Demonstrate Python Web Application Framework - Django with the suitable example. [8]
b) Use the knowledge of Cloud Computing to demonstrate
i) Amazon Auto Scaling.
ii) Xively Cloud for IoT. [10]

OR

- Q6)** a) Show that WAMP and its key concepts are useful in Cloud based IoT application Development. [8]
b) Apply the concept of cloud computing to design the smart irrigation system with proper explanation. [10]
- Q7)** a) Predict the possible challenges in designing secure IoT applications. [8]
b) Illustrate the classic pillars of information assurance while securing the IoT application. [9]

OR

- Q8)** a) Illustrate the threat model in securing IoT applications. [8]
b) Use security concepts to identify different threats (at least 03 in each) in the following IoT applications:
i) Smart Home Automation
ii) Smart Parking System
iii) Smart Irrigation System [9]

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